PROGRAM 8 – DEPTH FIRST SEARCH (DFS)

AIM

To write a Python program to implement the Depth First Search algorithm to traverse a graph.

ALGORITHM

1. Start the program.
2. Create a graph using an adjacency list.
3. Read the starting node.
4. Initialize an empty visited set.
5. Visit the starting node.
6. Mark the node as visited.
7. Display the current node.
8. Visit an unvisited adjacent node recursively.
9. Continue until all reachable nodes are visited.
10. Display the DFS traversal order.

CODE

```
graph={"A":["B","C"],"B":["D","E"],"C":["F"],"D":[],"E":["F"],"F":[]}
visited=set()
def dfs(node):
    if node not in visited:
        print(node,end=" ")
        visited.add(node)
        for neighbor in graph[node]:
            dfs(neighbor)
start=input("Enter starting node: ").upper()
if start in graph:
    print("DFS Traversal:")
    dfs(start)
else:
    print("Invalid starting node")
```

INPUT

```
Enter starting node: A
```

OUTPUT



RESULT

Thus, the Depth First Search algorithm was successfully implemented in Python to traverse the graph.